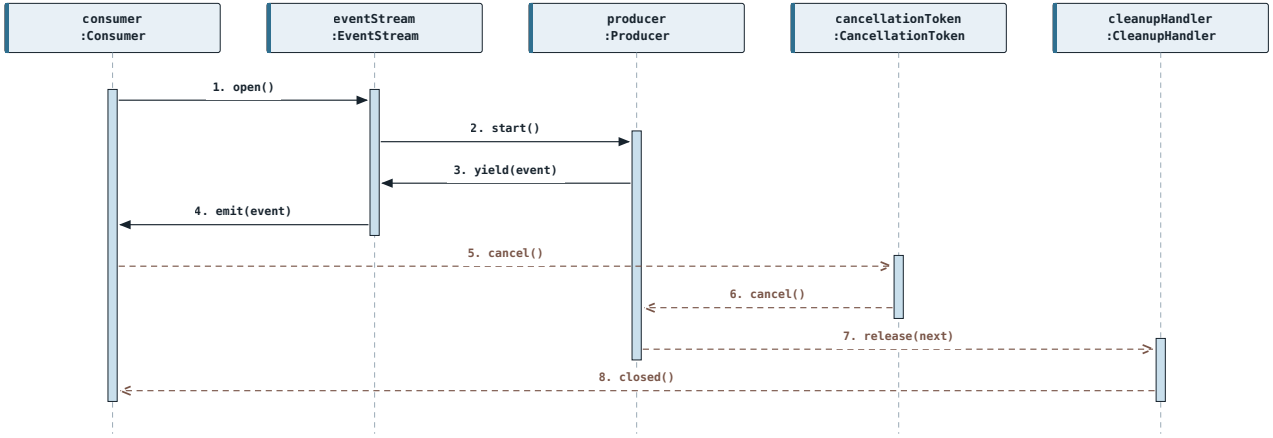
P05 Asynchronous-generator loop

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Expose incremental execution events.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A consumer observes an agent run incrementally.
TRIGGER	The runtime emits a semantic event or streamed token.
STRUCTURE	Async yield/resume, cancellation, and cleanup ownership.
EFFECT	Consumers receive ordered events and early exit releases runtime resources.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	The producer exposes an async iterator or equivalent streaming boundary.
ENSURES	Each emitted event has a stable type and sequence context.
INVARIANT	Cancellation and producer failure do not leave unreleased work.
VIOLATION	Abandoned consumers trigger cancellation or documented cleanup.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.